

TUGAS WEB SERVICE
DOKUMENTASI PENGUJIAN BLACK BOX TESTING
Sistem Manajemen Resep Masakan



Disusun oleh :

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PROGRAM STUDI TEKNIK INFORMATIKA S-1
FAKULTAS TEKNOLOGI INDUSTRI
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1. Pendahuluan

Dokumen ini merupakan laporan hasil pengujian Black Box Testing terhadap REST API Sistem Manajemen Resep Masakan. Pengujian dilakukan untuk memverifikasi bahwa seluruh endpoint API berfungsi sesuai spesifikasi, baik pada skenario positif (input valid) maupun skenario negatif (input tidak valid / kondisi error).

2. Informasi Sistem

Nama Sistem	Sistem Manajemen Resep Masakan API
Framework	Laravel 11
Database	MySQL
Autentikasi	Laravel Sanctum (Bearer Token)
Base URL	http://localhost:8000/api
Format Request	application/json / multipart/form-data
Alat Pengujian	Postman
Tanggal Pengujian	15 Juni 2026

3. Metode Pengujian

Metode yang digunakan adalah Black Box Testing, yaitu pengujian fungsional tanpa melihat kode internal. Setiap endpoint diuji dengan mengirimkan request melalui Postman dan memverifikasi response yang dikembalikan server.

Kriteria penilaian:

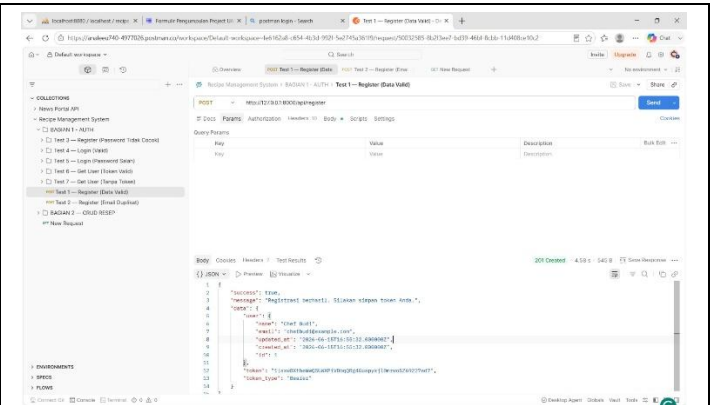
- PASS: Response HTTP status dan body sesuai dengan expected result
- FAIL: Response tidak sesuai expected result

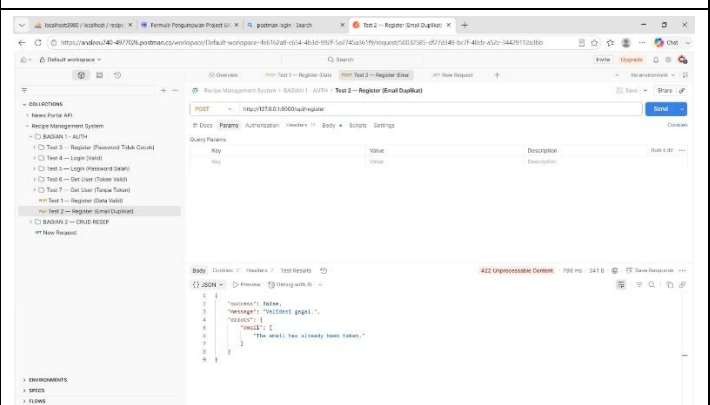
4. Hasil Pengujian

4.1 Pengujian Autentikasi (Auth)

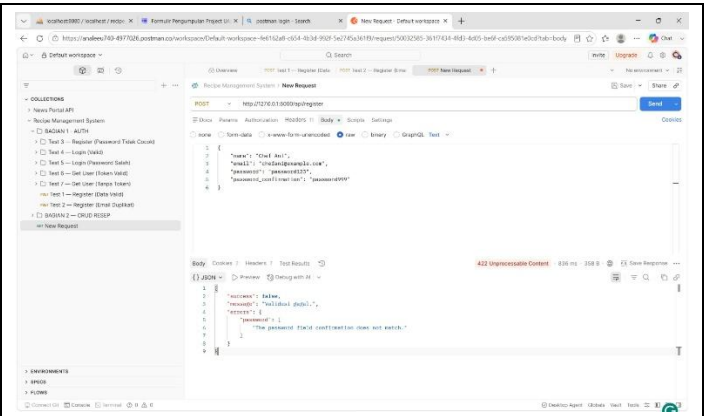
Mencakup endpoint Register, Login, Get User, dan Logout.

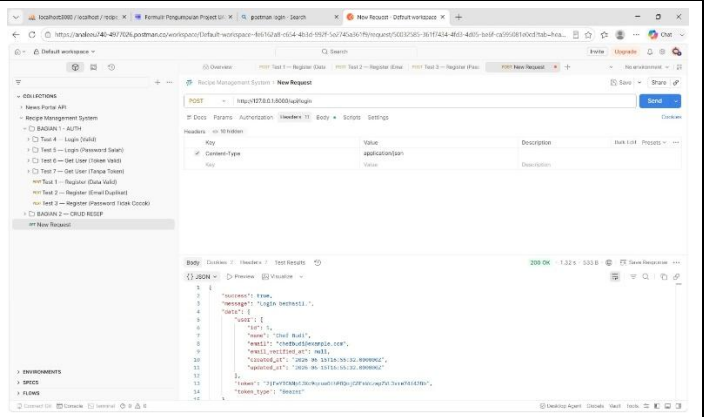
Test Case 1 Register - Data Valid	
Endpoint	[POST] /api/register
Input / Kondisi	name: "Chef Budi", email: "chefbudi@example.com", password: "password123", password_confirmation: "password123"
Expected Result	HTTP 201 - Registrasi berhasil, mengembalikan data user dan access token
Status	• PASS

Screenshot	 The screenshot shows a Postman interface with a collection named 'Recipe Management System'. The selected test is 'Test 1 - Register (Data Valid)'. The request is a POST to 'http://127.0.0.1:8080/api/register'. The response is a 201 Created status with a JSON body: { "success": true, "message": "Registration successful. Please check your email.", "token": { "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1bm90cm9scy5kaW5kaW9uIjoiIiwiaWF0IjoiMTY1OTU1MjU0LjE1In0.eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1bm90cm9scy5kaW5kaW9uIjoiIiwiaWF0IjoiMTY1OTU1MjU0LjE1In0", "token_type": "Bearer" } }.
(Test Case 1 - Register - Data Valid)	

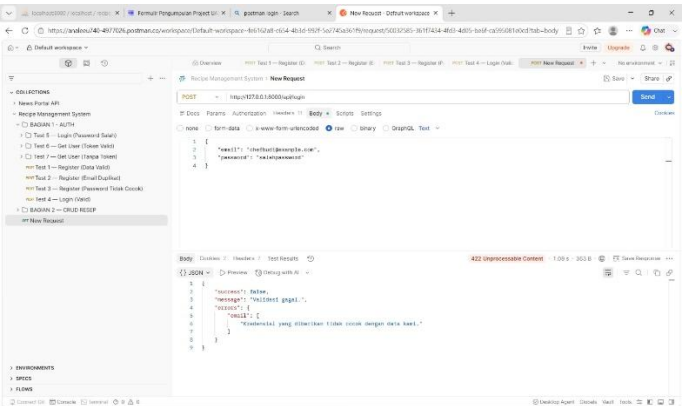
Test Case 2 Register - Email Duplikat	
Endpoint	[POST] /api/register
Input / Kondisi	Email yang sudah terdaftar sebelumnya
Expected Result	HTTP 422 - Error validasi: email sudah digunakan
Status	● PASS
Screenshot	 The screenshot shows a Postman interface with a collection named 'Recipe Management System'. The selected test is 'Test 2 - Register (Email Duplikat)'. The request is a POST to 'http://127.0.0.1:8080/api/register'. The response is a 422 Unprocessable Entity status with a JSON body: { "success": false, "message": "Registration failed.", "error": { "code": "EMAIL_ALREADY_EXISTS", "message": "The email has already been taken." } }.
(Test Case 2 - Register - Email Duplikat)	

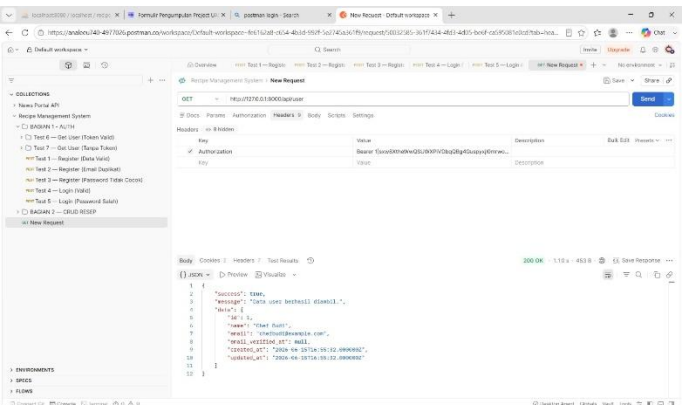
Test Case 3 Register - Password Tidak Cocok	
Endpoint	[POST] /api/register
Input / Kondisi	password: "password123", password_confirmation: "password999"
Expected Result	HTTP 422 - Error validasi: konfirmasi password tidak cocok
Status	● PASS

Screenshot	 (Test Case 3 - Register - Password Tidak Cocok)
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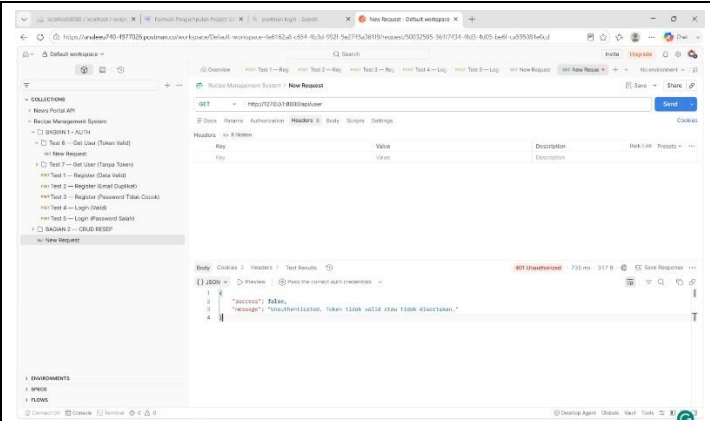
Test Case 4 Login - Kredensial Valid	
Endpoint	[POST] /api/login
Input / Kondisi	email: "chefbudi@example.com", password: "password123"
Expected Result	HTTP 200 - Login berhasil, mengembalikan data user dan access token
Status	● PASS
Screenshot	 (Test Case 4 - Login - Kredensial Valid)

Test Case 5 Login - Password Salah	
Endpoint	[POST] /api/login
Input / Kondisi	email: "chefbudi@example.com", password: "passwordsalah"
Expected Result	HTTP 422 - Error: kredensial tidak cocok dengan data kami
Status	● PASS

Screenshot	 (Test Case 5 - Login - Password Salah)
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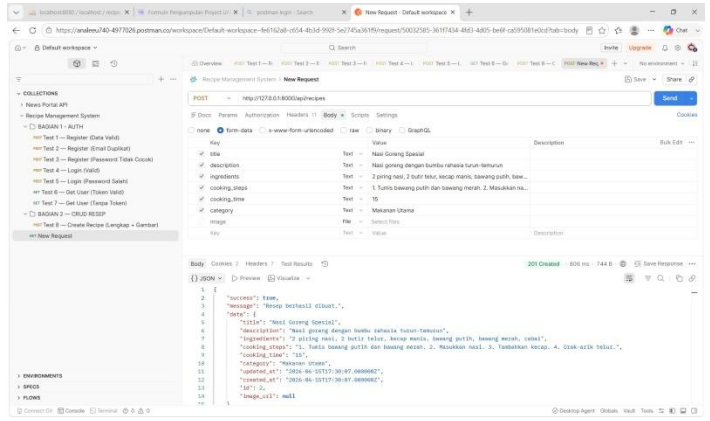
Test Case 6 Get User - Token Valid	
Endpoint	[GET] /api/user
Input / Kondisi	Header Authorization: Bearer <token valid>
Expected Result	HTTP 200 - Mengembalikan data profil user yang sedang login
Status	● PASS
Screenshot	 (Test Case 6 - Get User - Token Valid)

Test Case 7 Get User - Tanpa Token	
Endpoint	[GET] /api/user
Input / Kondisi	Tanpa header Authorization
Expected Result	HTTP 401 - Unauthenticated
Status	● PASS

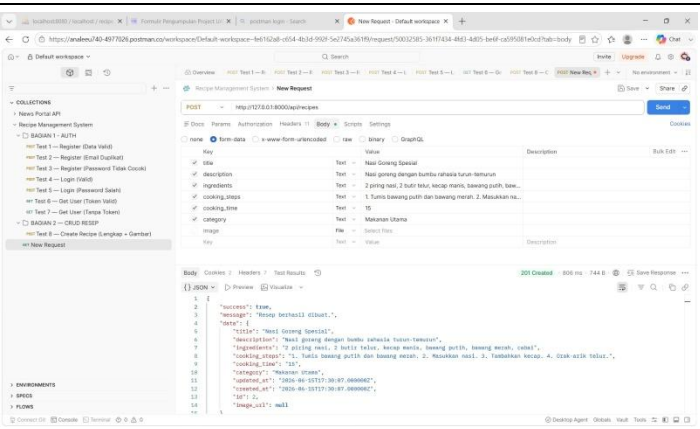
Screenshot	 (Test Case 7 - Get User - Tanpa Token)
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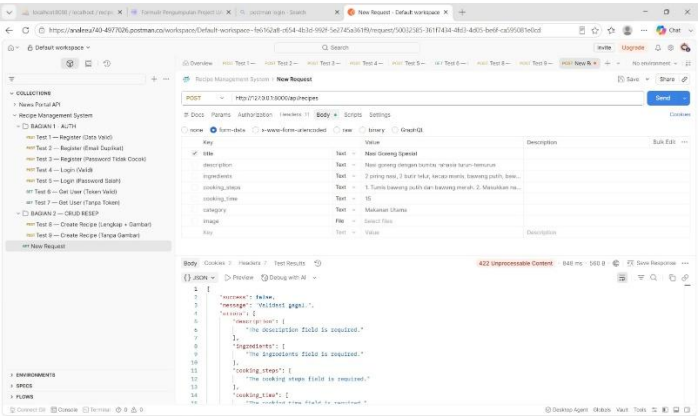
4.2 Pengujian CRUD Resep Masakan

Mencakup operasi Create, Read, Update, dan Delete pada data resep beserta validasi input.

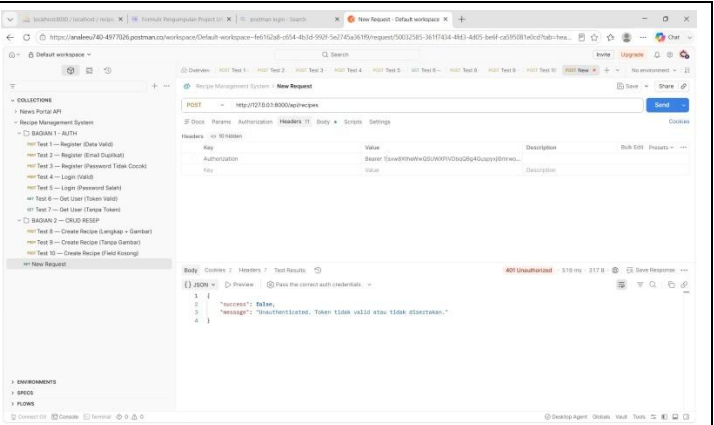
Test Case 8 Create Recipe - Tanpa Gambar	
Endpoint	[POST] /api/recipes
Input / Kondisi	Semua field teks diisi lengkap, field image dikosongkan
Expected Result	HTTP 201 - Resep berhasil dibuat, image_url bernilai null
Status	● PASS
Screenshot	 (Test Case 8 - Create Recipe - Tanpa Gambar)
Catatan	<i>Field image bersifat opsional (nullable), sehingga resep tetap berhasil dibuat tanpa gambar</i>

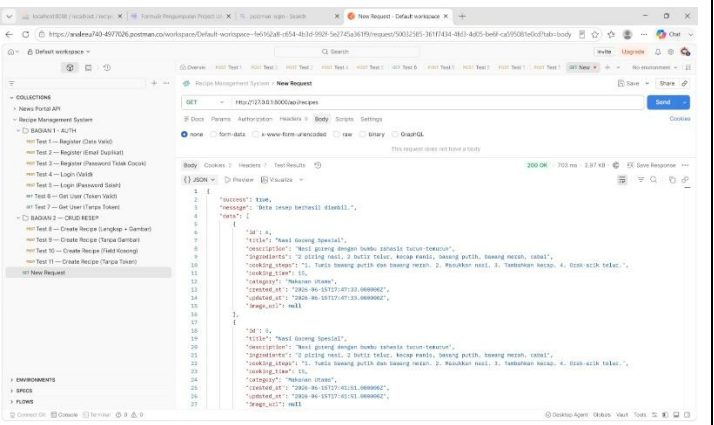
Test Case 9 Create Recipe - Field Wajib Kosong	
Endpoint	[POST] /api/recipes
Input / Kondisi	Hanya mengisi field title, field lain dikosongkan

Expected Result	HTTP 422 - Error validasi pada field yang kosong
Status	● PASS
Screenshot	 (Test Case 9 - Create Recipe - Field Wajib Kosong)

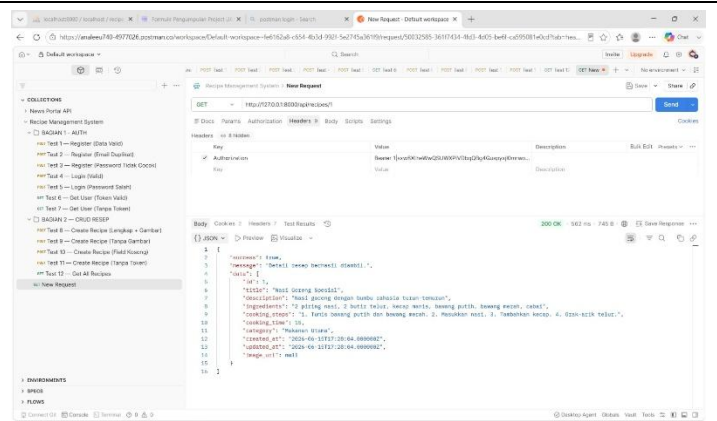
Test Case 10	Create Recipe - Tanpa Token
Endpoint	[POST] /api/recipes
Input / Kondisi	Tanpa header Authorization, body diisi lengkap
Expected Result	HTTP 401 - Unauthenticated
Status	● PASS
Screenshot	 (Test Case 10 - Create Recipe - Tanpa Token)

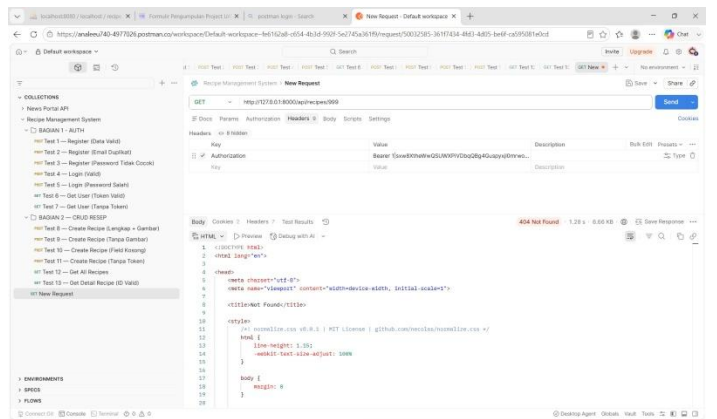
Test Case 11	Get All Recipes - Token Valid
Endpoint	[GET] /api/recipes
Input / Kondisi	Header Authorization: Bearer <token valid>
Expected Result	HTTP 200 - Mengembalikan array daftar semua resep
Status	● PASS

Screenshot	 (Test Case 11 - Get All Recipes - Token Valid)
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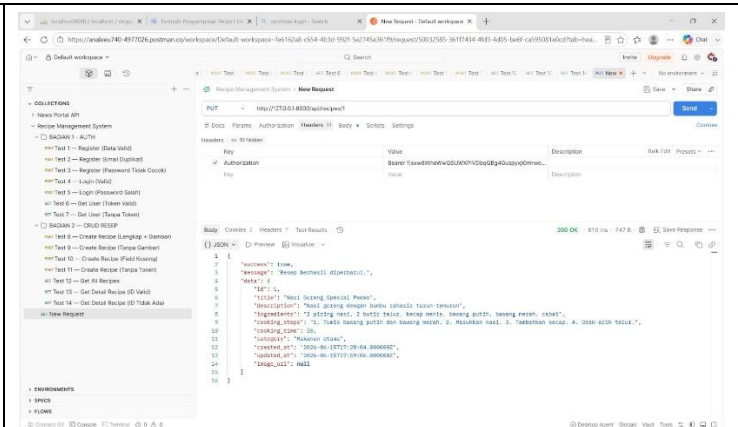
Test Case 12	Get All Recipes - Tanpa Token
Endpoint	[GET] /api/recipes
Input / Kondisi	Tanpa header Authorization
Expected Result	HTTP 401 - Unauthenticated
Status	● PASS
Screenshot	 (Test Case 12 - Get All Recipes - Tanpa Token)

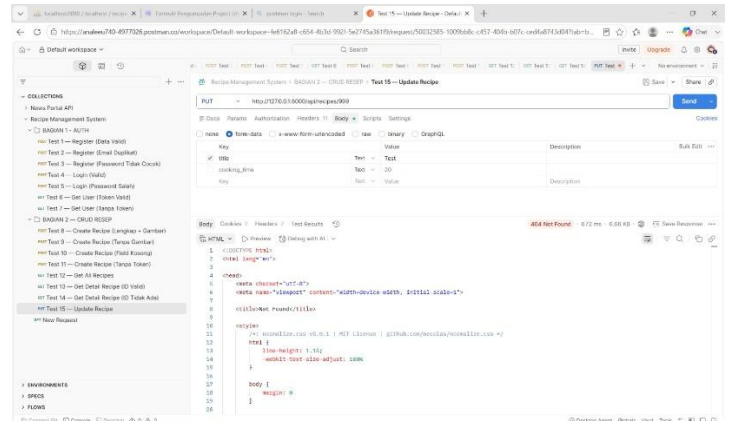
Test Case 13	Get Detail Recipe - ID Valid
Endpoint	[GET] /api/recipes/{id}
Input / Kondisi	ID resep yang ada di database, token valid
Expected Result	HTTP 200 - Mengembalikan detail satu resep sesuai ID
Status	● PASS

Screenshot	 (Test Case 13 - Get Detail Recipe - ID Valid)
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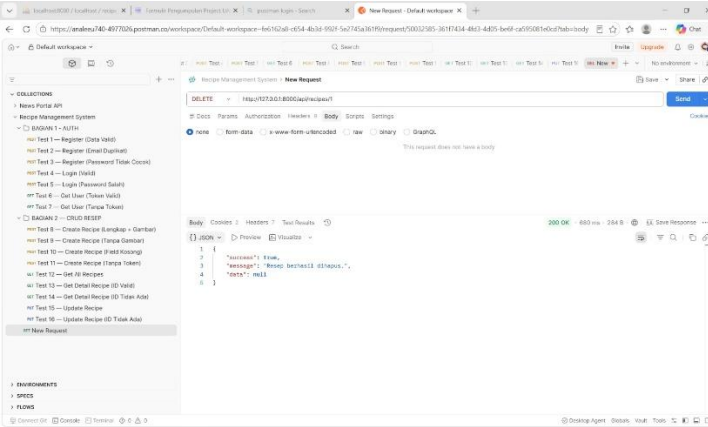
Test Case 14	Get Detail Recipe - ID Tidak Ada
Endpoint	[GET] /api/recipes/{id}
Input / Kondisi	ID = 999 (tidak ada di database), token valid
Expected Result	HTTP 404 - Data tidak ditemukan
Status	● PASS
Screenshot	 (Test Case 14 - Get Detail Recipe - ID Tidak Ada)

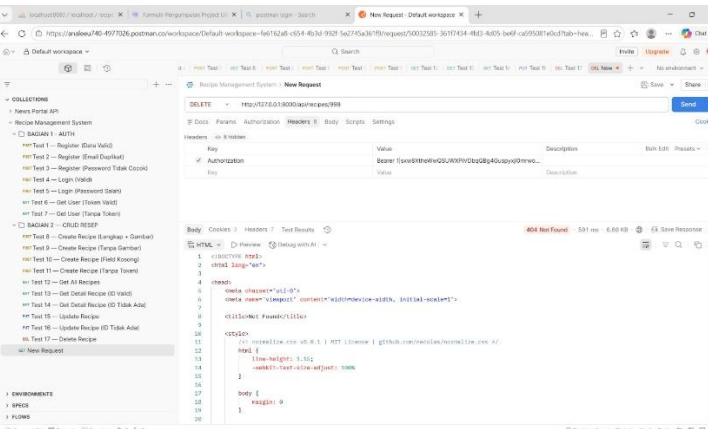
Test Case 15	Update Recipe - Data Valid
Endpoint	[PUT] /api/recipes/{id}
Input / Kondisi	ID valid, mengubah field title dan cooking_time, token valid
Expected Result	HTTP 200 - Data resep berhasil diperbarui
Status	● PASS

Screenshot	 (Test Case 15 - Update Recipe - Data Valid)
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Test Case 16 Update Recipe - ID Tidak Ada	
Endpoint	[PUT] /api/recipes/{id}
Input / Kondisi	ID = 999, body diisi, token valid
Expected Result	HTTP 404 - Data tidak ditemukan
Status	● PASS
Screenshot	 (Test Case 16 - Update Recipe - ID Tidak Ada)

Test Case 17 Delete Recipe - ID Valid	
Endpoint	[DELETE] /api/recipes/{id}
Input / Kondisi	ID resep yang ada di database, token valid
Expected Result	HTTP 200 - Resep berhasil dihapus dari database
Status	● PASS

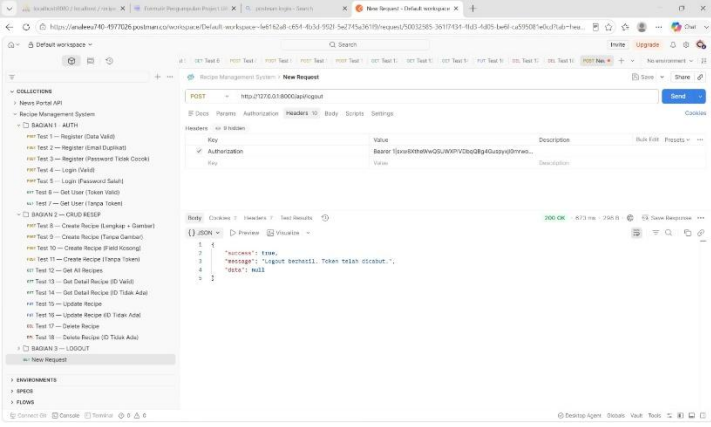
Screenshot	 (Test Case 17 - Delete Recipe - ID Valid)
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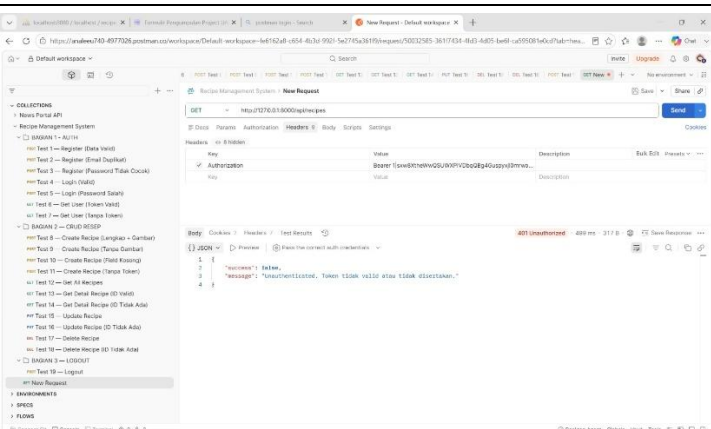
Test Case 18 Delete Recipe - ID Tidak Ada	
Endpoint	[DELETE] /api/recipes/{id}
Input / Kondisi	ID = 999, token valid
Expected Result	HTTP 404 - Data tidak ditemukan
Status	● PASS
Screenshot	 (Test Case 18 - Delete Recipe - ID Tidak Ada)

4.3 Pengujian Logout & Invalidasi Token

Memastikan token tidak bisa digunakan setelah logout.

Test Case 19 Logout - Token Valid	
Endpoint	[POST] /api/logout
Input / Kondisi	Header Authorization: Bearer <token valid>
Expected Result	HTTP 200 - Logout berhasil, token dicabut

Status	● PASS
Screenshot	 (Test Case 19 - Logout - Token Valid)

Test Case 20 Akses Setelah Logout	
Endpoint	[GET] /api/recipes
Input / Kondisi	Menggunakan token yang sudah di-logout
Expected Result	HTTP 401 - Unauthenticated (token tidak berlaku lagi)
Status	● PASS
Screenshot	 (Test Case 20 - Akses Setelah Logout)

5. Rekap Hasil Pengujian

No	Skenario Uji	Endpoint	Method	Input	Expected Result	Status
1	Register - Data Valid	/api/register	POST	name: Chef Budi...	HTTP 201	PASS
2	Register - Email Duplikat	/api/register	POST	Email yang sudah terdaftar sebelumnya...	HTTP 201	PASS

3	Register - Password Tidak Cocok	/api/register	POST	password: password123...	HTTP 201	PASS
4	Login - Kredensial Valid	/api/login	POST	email: chefbudi@example.com...	HTTP 200	PASS
5	Login - Password Salah	/api/login	POST	email: chefbudi@example.com...	HTTP 200	PASS
6	Get User - Token Valid	/api/user	GET	Header Authorization: Bearer <token vali...	HTTP 200	PASS
7	Get User - Tanpa Token	/api/user	GET	Tanpa header Authorization...	HTTP 200	PASS
8	Create Recipe - Tanpa Gambar	/api/recipes	POST	Semua field teks diisi lengkap...	HTTP 200	PASS
9	Create Recipe - Field Wajib Kosong	/api/recipes	POST	Hanya mengisi field title...	HTTP 200	PASS
10	Create Recipe - Tanpa Token	/api/recipes	POST	Tanpa header Authorization...	HTTP 200	PASS
11	Get All Recipes - Token Valid	/api/recipes	GET	Header Authorization: Bearer <token vali...	HTTP 200	PASS
12	Get All Recipes - Tanpa Token	/api/recipes	GET	Tanpa header Authorization...	HTTP 200	PASS
13	Get Detail Recipe - ID Valid	/api/recipes/{id}	GET	ID resep yang ada di database...	HTTP 200	PASS

14	Get Detail Recipe - ID Tidak Ada	/api/recipes/{id}	GET	ID = 999 (tidak ada di database)...	HTTP 200	PASS
15	Update Recipe - Data Valid	/api/recipes/{id}	PUT	ID valid...	HTTP 200	PASS
16	Update Recipe - ID Tidak Ada	/api/recipes/{id}	PUT	ID = 999...	HTTP 200	PASS
17	Delete Recipe - ID Valid	/api/recipes/{id}	DELETE	ID resep yang ada di database...	HTTP 200	PASS
18	Delete Recipe - ID Tidak Ada	/api/recipes/{id}	DELETE	ID = 999...	HTTP 200	PASS
19	Logout - Token Valid	/api/logout	POST	Header Authorization: Bearer <token vali...	HTTP 200	PASS
20	Akses Setelah Logout	/api/recipes	GET	Menggunakan token yang sudah di-logout...	HTTP 200	PASS

6. Kesimpulan

Berdasarkan hasil pengujian Black Box Testing terhadap 20 test case, seluruh endpoint API Sistem Manajemen Resep Masakan berfungsi sesuai spesifikasi yang telah ditentukan dengan tingkat keberhasilan 100% (20/20 PASS).

Sistem berhasil menangani:

1. Proses autentikasi (Register, Login, Logout) dengan validasi yang tepat
2. Perlindungan endpoint menggunakan Bearer Token (Laravel Sanctum)
3. Operasi CRUD resep masakan dengan validasi input yang lengkap
4. Penanganan error yang konsisten (HTTP 401, 404, 422)
5. Invalidasi token setelah proses logout

Sistem dinyatakan layak dan siap digunakan sesuai dengan kebutuhan fungsional yang telah ditetapkan.